

- 7.2
1) Use an addition or subtraction formula to find the exact value of the expression.

$$\begin{aligned}\cos(165^\circ) &= \cos(120^\circ + 45^\circ) \\ &= \cos(120^\circ)\cos(45^\circ) - \sin(120^\circ)\sin(45^\circ) \\ &= \left(-\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)\end{aligned}$$

- 2) Use an addition or subtraction formula to find the exact value of the expression.

$$\begin{aligned}\sin\left(\frac{43\pi}{12}\right) &= \sin\left(\frac{19\pi}{12}\right) \\ &= \sin\left(\frac{3\pi}{12} + \frac{16\pi}{12}\right) \\ &= \sin\left(\frac{\pi}{4} + \frac{4\pi}{3}\right) \\ &= \sin\left(\frac{\pi}{4}\right)\cos\left(\frac{4\pi}{3}\right) + \cos\left(\frac{\pi}{4}\right)\sin\left(\frac{4\pi}{3}\right) \\ &= \left(\frac{\sqrt{2}}{2}\right)\left(-\frac{1}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(-\frac{\sqrt{3}}{2}\right)\end{aligned}$$

- 3) Use an addition or subtraction formula to find the exact value of the expression.

$$\begin{aligned}\tan\left(-\frac{11\pi}{12}\right) &= -\tan\left(\frac{11\pi}{12}\right) \\ &= -\tan\left(\frac{2\pi}{12} + \frac{9\pi}{12}\right) \\ &= -\tan\left(\frac{\pi}{6} + \frac{3\pi}{4}\right) \\ &= -\left(\frac{\tan\frac{\pi}{6} + \tan\frac{3\pi}{4}}{1 - \tan\frac{\pi}{6}\tan\frac{3\pi}{4}}\right) \\ &= -\left(\frac{\frac{1}{\sqrt{3}} + (-1)}{1 - \left(\frac{1}{\sqrt{3}}\right)(-1)}\right)\end{aligned}$$

- 4) Use an Addition or Subtraction Formula to write the expression as a trig function of one number.

$$\sin\left(\frac{4\pi}{5}\right)\cos\left(\frac{7\pi}{15}\right) - \cos\left(\frac{4\pi}{5}\right)\sin\left(\frac{7\pi}{15}\right)$$

$$s = \frac{4\pi}{5}, t = \frac{7\pi}{15}$$

$$\begin{aligned}\sin(s-t) &= \sin s \cos t - \cos s \sin t \\ &= \sin\left(\frac{4\pi}{5} - \frac{7\pi}{15}\right) \\ &= \sin\left(\frac{\pi}{3}\right)\end{aligned}$$

Find the exact value.

$$\begin{aligned}\sin\left(\frac{\pi}{3}\right) &= \frac{\sqrt{3}}{2}\end{aligned}$$

- 5) Use an Addition or Subtraction Formula to write the expression as a trig function of one number.

$$\frac{\tan(74^\circ) - \tan(14^\circ)}{1 + \tan(74^\circ)\tan(14^\circ)}$$

$$s = 74^\circ, t = 14^\circ$$

$$\begin{aligned}\tan(s-t) &= \frac{\tan s - \tan t}{1 + \tan s \tan t} \\ &= \tan(74^\circ - 14^\circ) \\ &= \tan(60^\circ)\end{aligned}$$

Find the exact value.

$$\begin{aligned}\tan(60^\circ) &= \sqrt{3}\end{aligned}$$

- 6) Prove the cofunction identity using the Addition and Subtraction Formulas for sine and cosine.

$$\cot\left(\frac{\pi}{2} - u\right) = \tan(u)$$

Select the correct identities, formulas, and algebraic manipulations that justify each step in the proof below.

$$\begin{aligned}\cot\left(\frac{\pi}{2} - u\right) &= \frac{\cos\left(\frac{\pi}{2} - u\right)}{\sin\left(\frac{\pi}{2} - u\right)} && \text{Reciprocal identity} \quad \checkmark \\ &= \frac{\cos\left(\frac{\pi}{2}\right)\cos(u) + \sin\left(\frac{\pi}{2}\right)\sin(u)}{\sin\left(\frac{\pi}{2}\right)\cos(u) - \cos\left(\frac{\pi}{2}\right)\sin(u)} && \text{Subtraction formulas for Sine and Cosine} \quad \checkmark \\ &= \frac{0 \cdot \cos(u) + 1 \cdot \sin(u)}{1 \cdot \cos(u) - 0 \cdot \sin(u)} && \text{Evaluate } \sin(\pi/2) \text{ and } \cos(\pi/2) \quad \checkmark \\ &= \frac{\sin(u)}{\cos(u)} && \text{Simplify} \quad \checkmark \\ &= \tan(u) && \text{Reciprocal identity} \quad \checkmark\end{aligned}$$

7) Prove the identity. (Simplify your answers completely.)

$$\sin\left(\frac{\pi}{2} - x\right) = \sin\left(\frac{\pi}{2} + x\right)$$

First, use the Subtraction Formula for Sine on the left-hand side of the identity, and simplify.

$$\begin{aligned} \text{LHS} &= \sin\left(\frac{\pi}{2} - x\right) \\ &= \sin\left(\frac{\pi}{2}\right)\left(\boxed{\cos(x)}\right) - \cos\left(\frac{\pi}{2}\right)\sin(x) \\ &= 1 \cdot \boxed{\cos(x)} - 0 \cdot \sin(x) \\ &= \boxed{\cos(x)} \quad \text{Excellent!} \end{aligned}$$

Next, use the Addition Formula for Sine on the right-hand side of the identity, and simplify.

$$\begin{aligned} \text{RHS} &= \sin\left(\frac{\pi}{2} + x\right) \\ &= \sin\left(\frac{\pi}{2}\right)\left(\boxed{\cos(x)}\right) + \cos\left(\frac{\pi}{2}\right)\sin(x) \\ &= 1 \cdot \boxed{\cos(x)} + 0 \cdot \sin(x) \\ &= \boxed{\cos(x)} \quad \text{Nice!} \end{aligned}$$

Because LHS = RHS, we have proven that $\sin\left(\frac{\pi}{2} - x\right) = \sin\left(\frac{\pi}{2} + x\right)$.

8) Prove the identity. (Simplify your answers completely.)

$$5 \cos(x + y) + 5 \cos(x - y) = 10 \cos(x) \cos(y)$$

$$\begin{aligned} 5 \cos(x + y) + 5 \cos(x - y) &= 5(\cos(x) \cos(y) - \sin(x) \sin(y)) + 5(\cos(x) \cos(y) + \sin(x) \sin(y)) \\ &= 5 \cos(x) \cos(y) - 5 \sin(x) \sin(y) + 5 \cos(x) \cos(y) + 5 \sin(x) \sin(y) \\ &= \boxed{10 \cos(x) \cos(y)} \quad \text{Good job.} \end{aligned}$$

9) Write the given expression in terms of x and y only.

$$\tan(\underbrace{\sin^{-1}(x)}_s) + \underbrace{\cos^{-1}(y)}_t$$

$$= \tan(s+t) = \frac{\tan s + \tan t}{1 - \tan s \tan t}$$

$$\begin{array}{c} x \\ \swarrow \quad \searrow \\ \sqrt{1-x^2} \end{array} \quad \begin{array}{c} \sqrt{1-y^2} \\ \swarrow \quad \searrow \\ y \end{array}$$

$$= \frac{\frac{x}{\sqrt{1-x^2}} + \frac{\sqrt{1-y^2}}{y}}{1 - \left(\frac{x}{\sqrt{1-x^2}}\right)\left(\frac{\sqrt{1-y^2}}{y}\right)}$$

10) Write the given expression in terms of x and y only.

$$\sin(\underbrace{\cot^{-1}(x)}_s) - \underbrace{\cot^{-1}(y)}_t$$

$$= \sin(s-t) = \sin s \cos t - \cos s \sin t$$

$$\begin{array}{c} 1 \\ \swarrow \quad \searrow \\ \sqrt{x^2+1} \quad x \end{array} \quad \begin{array}{c} \sqrt{y^2+1} \\ \swarrow \quad \searrow \\ y \end{array}$$

$$= \left(\frac{1}{\sqrt{x^2+1}}\right)\left(\frac{y}{\sqrt{y^2+1}}\right) - \left(\frac{x}{\sqrt{x^2+1}}\right)\left(\frac{1}{\sqrt{y^2+1}}\right)$$

11) Find the exact value of the expression

$$\sin(\underbrace{\cos^{-1}\left(\frac{3}{4}\right)}_s) - \underbrace{\tan^{-1}\left(\frac{1}{3}\right)}_t$$

$$= \sin(s-t) = \sin s \cos t - \cos s \sin t$$

$$\begin{array}{c} \sqrt{7} \quad 4 \\ \swarrow \quad \searrow \\ 3 \end{array} \quad \begin{array}{c} 1 \quad \sqrt{10} \\ \swarrow \quad \searrow \\ 3 \end{array}$$

$$= \left(\frac{\sqrt{7}}{4}\right)\left(\frac{3}{\sqrt{10}}\right) - \left(\frac{3}{4}\right)\left(\frac{1}{\sqrt{10}}\right)$$

12) Evaluate expression under given conditions.

$$\cos(\theta - \phi); \cos(\theta) = \frac{5}{13}, \theta \text{ in } \text{QIV}, \tan(\phi) = -\sqrt{3}, \phi \text{ in } \text{QII}$$

$$\begin{array}{c} \theta \text{ QIV} \\ | \\ \star \\ -12 \quad \sqrt{13} \end{array} \quad \begin{array}{c} \phi \text{ QII} \\ | \\ \star \\ 2 \quad \sqrt{3} \end{array}$$

$$\begin{aligned} \cos(\theta - \phi) &= \cos \theta \cos \phi + \sin \theta \sin \phi \\ &= \left(\frac{5}{13}\right)\left(-\frac{1}{2}\right) + \left(-\frac{12}{13}\right)\left(\frac{\sqrt{3}}{2}\right) \end{aligned}$$

13) Write the expression in terms of sine only.

$$\sin(9x) - \cos(9x)$$

$$A = 1, B = -1$$

$$K = \sqrt{1^2 + (-1)^2}$$

$$= \sqrt{2}$$

$$\cos \Phi = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\sin \Phi = -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$$

$$\Phi = \frac{7\pi}{4}$$

$$\sin(9x) - \cos(9x) = \sqrt{2} \sin\left(9x + \frac{7\pi}{4}\right)$$

14) Write the expression in terms of sine only.

$$3 \sin(3\pi x) + 3\sqrt{3} \cos(3\pi x)$$

$$A = 3, B = 3\sqrt{3}$$

$$K = \sqrt{3^2 + (3\sqrt{3})^2}$$

$$= 6$$

$$\cos \Phi = \frac{3}{6} = \frac{1}{2}$$

$$\sin \Phi = \frac{3\sqrt{3}}{6} = \frac{\sqrt{3}}{2}$$

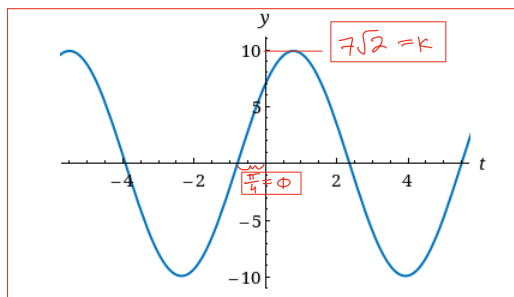
$$\Phi = \frac{\pi}{3}$$

$$3 \sin(3\pi x) + 3\sqrt{3} \cos(3\pi x) = 6 \sin\left(3\pi x + \frac{\pi}{3}\right)$$

15) A graphing device is recommended.

A digital delay device echoes an input signal by repeating it a fixed length of time after it is received. If such a device receives the pure note $f_1(t) = 7 \sin(t)$ and echoes the pure note $f_2(t) = 7 \cos(t)$, then the combined sound is $f(t) = f_1(t) + f_2(t)$.

(a) Graph $y = f(t)$ and observe that the graph has the form of a sine curve $y = k \sin(t + \varphi)$.



16) Use an Addition or Subtraction Formula to write the expression as a trig function of one number.

$$\sin(10^\circ) \cos(20^\circ) + \cos(10^\circ) \sin(20^\circ)$$

$$s = 10^\circ, t = 20^\circ$$

$$\sin(s+t) = \sin s \cos t + \cos s \sin t$$

$$= \sin(10^\circ + 20^\circ)$$

$$= \sin(30^\circ)$$

Find the exact value.

$$\sin(30^\circ)$$

$$= \frac{1}{2}$$

17) Use an Addition or Subtraction Formula to write the expression as a trig function of one number.

$$\cos\left(\frac{17\pi}{18}\right) \cos\left(\frac{\pi}{9}\right) + \sin\left(\frac{17\pi}{18}\right) \sin\left(\frac{\pi}{9}\right)$$

$$s = \frac{17\pi}{18}, t = \frac{\pi}{9}$$

$$\cos(s-t) = \cos s \cos t + \sin s \sin t$$

$$= \cos\left(\frac{17\pi}{18} - \frac{\pi}{9}\right)$$

$$= \cos\left(\frac{5\pi}{6}\right)$$

Find the exact value.

$$\cos\left(\frac{5\pi}{6}\right)$$

$$= -\frac{\sqrt{3}}{2}$$